CYBERSTER PURPLE TEAM

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Abstract

This report presents the design, implementation, and evaluation of Network Intrusion Detection and Prevention mechanisms using Suricata. The lab emphasizes the transition from passive

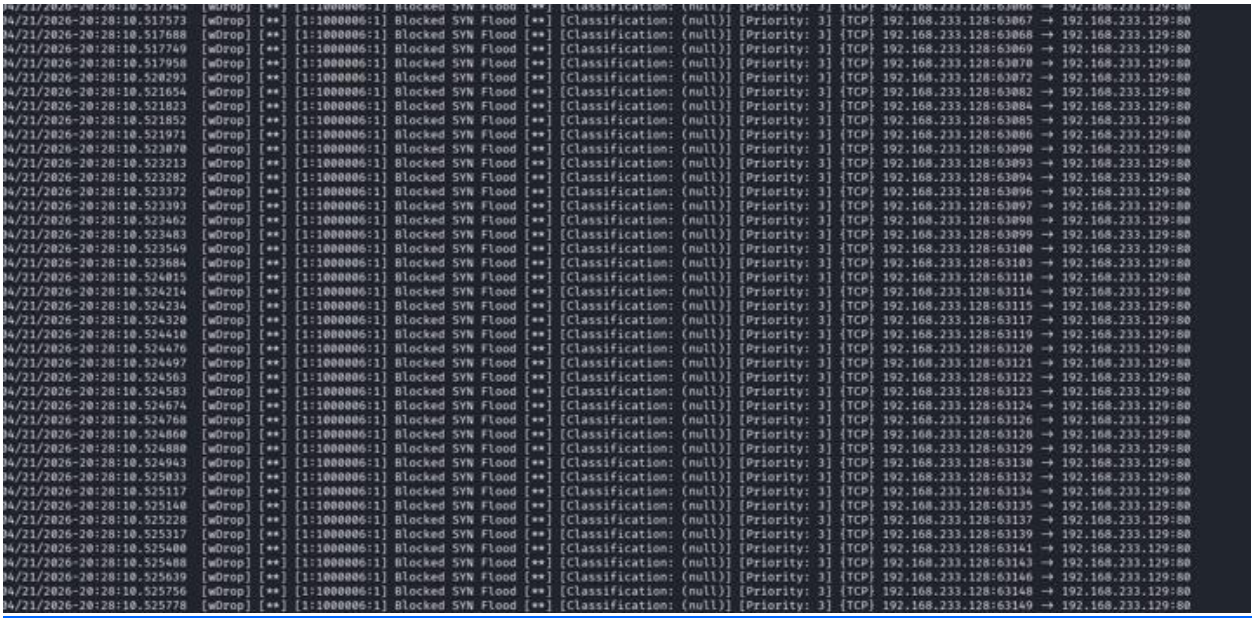
monitoring (IDS) to active threat mitigation (IPS), incorporating real-time detection rules, kernel-level packet inspection, and Layer 2 attack analysis.

1. Introduction

Modern cybersecurity strategies prioritize proactive defense mechanisms over passive monitoring. This lab focuses on deploying Suricata as a network defense engine to achieve deep packet inspection, threat detection, and prevention.

The primary objective was to:

- Establish network visibility
- Implement signature-based detection
- Transition from IDS to IPS for real-time threat mitigation



2. System Architecture

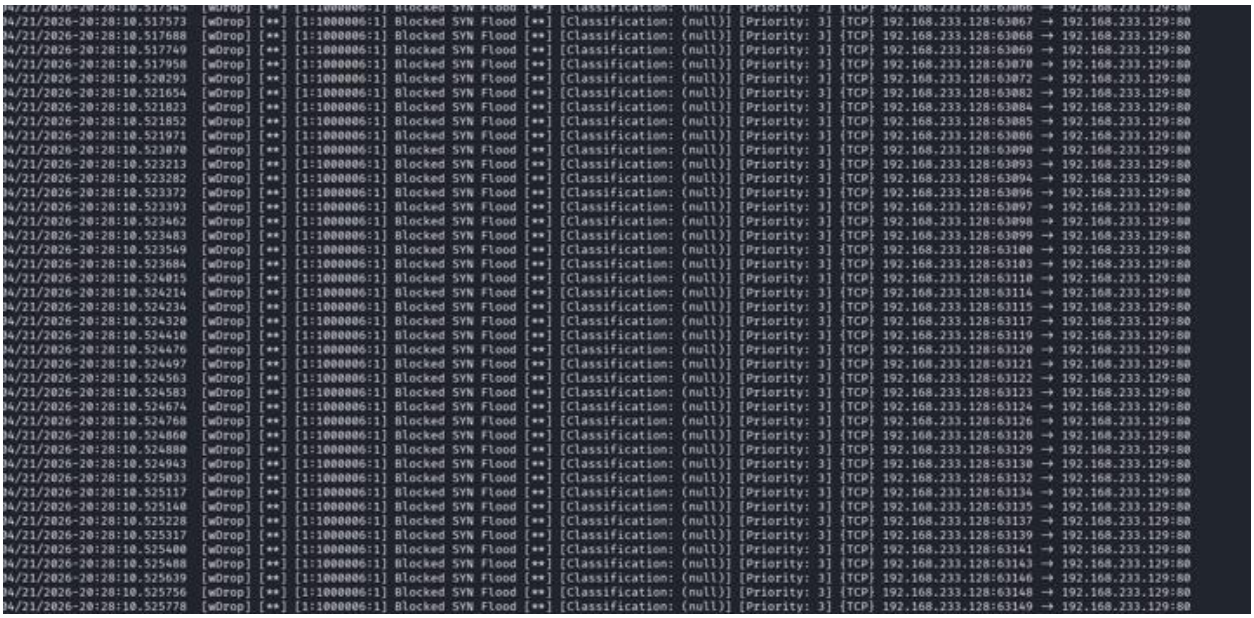
2.1 IDS vs. IPS Analysis

Feature	IDS (Intrusion Detection System)	IPS (Intrusion Prevention System)
Connectivity	Passive (SPAN/Mirror Port)	Active (In-Line)
Functionality	Detection & Alerting	Detection & Blocking
Latency	None (Traffic Copy)	Minimal (Processing Delay)
Output	Logs & Alerts	Dropped/Blocked Packets

2.2 Environment Setup

- **Platform:** Kali Linux
- **Detection Engine:** Suricata
- **Network Mode:** Promiscuous Mode

Promiscuous mode enables the system to capture all packets in the network segment, not just those addressed to the host, ensuring full traffic visibility.



```
04/21/2026-20:28:10.517573  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63067 → 192.168.233.129:80
04/21/2026-20:28:10.517688  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63068 → 192.168.233.129:80
04/21/2026-20:28:10.517749  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63069 → 192.168.233.129:80
04/21/2026-20:28:10.517958  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63070 → 192.168.233.129:80
04/21/2026-20:28:10.520293  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63072 → 192.168.233.129:80
04/21/2026-20:28:10.521654  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63082 → 192.168.233.129:80
04/21/2026-20:28:10.521823  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63084 → 192.168.233.129:80
04/21/2026-20:28:10.521852  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63085 → 192.168.233.129:80
04/21/2026-20:28:10.521971  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63086 → 192.168.233.129:80
04/21/2026-20:28:10.523070  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63090 → 192.168.233.129:80
04/21/2026-20:28:10.523213  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63093 → 192.168.233.129:80
04/21/2026-20:28:10.523282  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63094 → 192.168.233.129:80
04/21/2026-20:28:10.523372  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63096 → 192.168.233.129:80
04/21/2026-20:28:10.523393  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63097 → 192.168.233.129:80
04/21/2026-20:28:10.523462  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63098 → 192.168.233.129:80
04/21/2026-20:28:10.523483  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63099 → 192.168.233.129:80
04/21/2026-20:28:10.523549  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63100 → 192.168.233.129:80
04/21/2026-20:28:10.523684  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63103 → 192.168.233.129:80
04/21/2026-20:28:10.524015  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63110 → 192.168.233.129:80
04/21/2026-20:28:10.524214  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63114 → 192.168.233.129:80
04/21/2026-20:28:10.524234  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63115 → 192.168.233.129:80
04/21/2026-20:28:10.524320  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63117 → 192.168.233.129:80
04/21/2026-20:28:10.524410  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63119 → 192.168.233.129:80
04/21/2026-20:28:10.524476  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63120 → 192.168.233.129:80
04/21/2026-20:28:10.524497  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63121 → 192.168.233.129:80
04/21/2026-20:28:10.524563  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63122 → 192.168.233.129:80
04/21/2026-20:28:10.524583  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63123 → 192.168.233.129:80
04/21/2026-20:28:10.524674  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63124 → 192.168.233.129:80
04/21/2026-20:28:10.524768  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63126 → 192.168.233.129:80
04/21/2026-20:28:10.524860  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63128 → 192.168.233.129:80
04/21/2026-20:28:10.524880  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63129 → 192.168.233.129:80
04/21/2026-20:28:10.524943  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63130 → 192.168.233.129:80
04/21/2026-20:28:10.525033  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63132 → 192.168.233.129:80
04/21/2026-20:28:10.525117  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63134 → 192.168.233.129:80
04/21/2026-20:28:10.525140  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63135 → 192.168.233.129:80
04/21/2026-20:28:10.525228  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63137 → 192.168.233.129:80
04/21/2026-20:28:10.525317  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63139 → 192.168.233.129:80
04/21/2026-20:28:10.525400  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63141 → 192.168.233.129:80
04/21/2026-20:28:10.525488  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63143 → 192.168.233.129:80
04/21/2026-20:28:10.525639  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63146 → 192.168.233.129:80
04/21/2026-20:28:10.525756  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63148 → 192.168.233.129:80
04/21/2026-20:28:10.525778  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:63149 → 192.168.233.129:80
```

3. Detection Rule Implementation

3.1 ICMP Reconnaissance Detection

To detect early-stage reconnaissance activity, an ICMP rule was implemented:

```
alert icmp any any -> any any (msg:"ICMP Ping Detected"; sid:1000001; rev:1;)
```

Outcome:
This rule successfully identifies ping sweeps used for network discovery.

3.2 SYN Flood Detection (DoS Protection)

A rule was configured to detect high-frequency SYN packets:

```
alert tcp any any -> any 80 (msg:"SYN Flood Detected"; flags:S;
threshold:type both, track by_src, count 20, seconds 1; sid:1000004; rev:1;)
```

Analysis:

- Detects abnormal traffic spikes
- Uses thresholding to reduce false positives
- Differentiates between normal and malicious behavior

4. Intrusion Prevention System (IPS) Implementation

4.1 NFQUEUE Integration

To enable inline packet inspection, Suricata was integrated with the Linux firewall:

```
sudo iptables -I INPUT -j NFQUEUE --queue-num 0
```

Function:
Packets are redirected to Suricata before reaching the system, allowing inspection and decision-making.

```
04/21/2026-21:06:18.219654  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57569 → 192.168.233.129:80
04/21/2026-21:06:18.221231  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57571 → 192.168.233.129:80
04/21/2026-21:06:18.221635  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57573 → 192.168.233.129:80
04/21/2026-21:06:18.221987  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57575 → 192.168.233.129:80
04/21/2026-21:06:18.257882  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57576 → 192.168.233.129:80
04/21/2026-21:06:18.264213  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57579 → 192.168.233.129:80
04/21/2026-21:06:18.264595  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57580 → 192.168.233.129:80
04/21/2026-21:06:18.264674  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57581 → 192.168.233.129:80
04/21/2026-21:06:18.265162  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57584 → 192.168.233.129:80
04/21/2026-21:06:18.265465  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57587 → 192.168.233.129:80
04/21/2026-21:06:18.265595  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57589 → 192.168.233.129:80
04/21/2026-21:06:18.265698  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57590 → 192.168.233.129:80
04/21/2026-21:06:18.265729  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57591 → 192.168.233.129:80
04/21/2026-21:06:18.265862  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57593 → 192.168.233.129:80
04/21/2026-21:06:18.265950  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57594 → 192.168.233.129:80
04/21/2026-21:06:18.265976  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57595 → 192.168.233.129:80
04/21/2026-21:06:18.266560  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57598 → 192.168.233.129:80
04/21/2026-21:06:18.266592  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57599 → 192.168.233.129:80
04/21/2026-21:06:18.266689  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57600 → 192.168.233.129:80
04/21/2026-21:06:18.266821  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57602 → 192.168.233.129:80
04/21/2026-21:06:18.266946  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57604 → 192.168.233.129:80
04/21/2026-21:06:18.266981  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57605 → 192.168.233.129:80
04/21/2026-21:06:18.267289  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57609 → 192.168.233.129:80
04/21/2026-21:06:18.267519  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57612 → 192.168.233.129:80
04/21/2026-21:06:18.268173  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57616 → 192.168.233.129:80
04/21/2026-21:06:18.268403  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57618 → 192.168.233.129:80
04/21/2026-21:06:18.268440  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57619 → 192.168.233.129:80
04/21/2026-21:06:18.268807  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57622 → 192.168.233.129:80
04/21/2026-21:06:18.269142  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57624 → 192.168.233.129:80
04/21/2026-21:06:18.269296  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57626 → 192.168.233.129:80
04/21/2026-21:06:18.269324  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57627 → 192.168.233.129:80
04/21/2026-21:06:18.269530  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57630 → 192.168.233.129:80
04/21/2026-21:06:18.269646  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57632 → 192.168.233.129:80
04/21/2026-21:06:18.269790  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57634 → 192.168.233.129:80
04/21/2026-21:06:18.269816  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57635 → 192.168.233.129:80
04/21/2026-21:06:18.269900  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57636 → 192.168.233.129:80
04/21/2026-21:06:18.270041  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57639 → 192.168.233.129:80
04/21/2026-21:06:18.297924  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57640 → 192.168.233.129:80
04/21/2026-21:06:18.298320  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57641 → 192.168.233.129:80
04/21/2026-21:06:18.298402  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57643 → 192.168.233.129:80
04/21/2026-21:06:18.298477  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57645 → 192.168.233.129:80
04/21/2026-21:06:18.298510  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57646 → 192.168.233.129:80
04/21/2026-21:06:18.298521  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57647 → 192.168.233.129:80
04/21/2026-21:06:18.299001  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57648 → 192.168.233.129:80
04/21/2026-21:06:18.299015  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57649 → 192.168.233.129:80
04/21/2026-21:06:18.299443  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.16
```

4.2 Active Threat Mitigation

The detection rule was modified from:

```
alert → drop
```

Result:

- Malicious packets were blocked in real time

- Legitimate traffic remained unaffected
- Web server maintained availability during attack

5. Layer 2 Security: ARP Spoofing

Attack Simulation

- Tool Used: `arpspoof`
- Attack Type: Man-in-the-Middle (MITM)

Detection Strategy

- Monitor abnormal ARP responses
- Identify duplicate MAC-IP mappings
- Trace attacker source within LAN

6. Log Analysis & Findings

Log file: `/var/log/suricata/fast.log`

Observed Events

- **ICMP Alerts:**
  - Source: 192.168.233.128
  - Activity: Network reconnaissance
- **SYN Flood Prevention:**
  - Multiple `[Drop]` actions recorded
- **Normal Traffic Baseline:**
  - Standard TCP handshakes observed
  - Used for behavior comparison

7. Discussion

The transition from IDS to IPS significantly enhances network security by enabling automated responses. While IDS provides visibility, IPS ensures active defense against threats such as DoS attacks.

However, IPS introduces slight latency and requires careful tuning to avoid false positives.

```
04/21/2026-20:14:05.642906  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:43329 -> 192.168.233.129:80
04/21/2026-20:14:06.643523  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:1956 -> 192.168.233.129:80
04/21/2026-20:14:07.643469  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:28400 -> 192.168.233.129:80
04/21/2026-20:14:08.643995  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:53511 -> 192.168.233.129:80
04/21/2026-20:14:09.643377  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12999 -> 192.168.233.129:80
04/21/2026-20:14:10.643361  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:35919 -> 192.168.233.129:80
04/21/2026-20:14:11.643843  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:59204 -> 192.168.233.129:80
04/21/2026-20:14:12.643933  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:17348 -> 192.168.233.129:80
04/21/2026-20:14:13.644111  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:39475 -> 192.168.233.129:80
04/21/2026-20:14:14.645449  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:64369 -> 192.168.233.129:80
04/21/2026-20:14:15.645997  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:21575 -> 192.168.233.129:80
04/21/2026-20:14:16.643717  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:43090 -> 192.168.233.129:80
04/21/2026-20:14:17.644225  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:2049 -> 192.168.233.129:80
04/21/2026-20:14:18.644149  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:31944 -> 192.168.233.129:80
04/21/2026-20:14:19.644060  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:62701 -> 192.168.233.129:80
04/21/2026-20:14:20.643737  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:27069 -> 192.168.233.129:80
04/21/2026-20:14:21.644282  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:57720 -> 192.168.233.129:80
04/21/2026-20:14:22.644482  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:20306 -> 192.168.233.129:80
04/21/2026-20:14:23.644145  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:47414 -> 192.168.233.129:80
04/21/2026-20:14:24.644244  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:8566 -> 192.168.233.129:80
04/21/2026-20:14:25.644105  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:35515 -> 192.168.233.129:80
04/21/2026-20:14:26.644423  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:60802 -> 192.168.233.129:80
04/21/2026-20:14:27.644252  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:16522 -> 192.168.233.129:80
04/21/2026-20:14:28.645243  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:32777 -> 192.168.233.129:80
04/21/2026-20:14:29.645891  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:49350 -> 192.168.233.129:80
04/21/2026-20:14:30.646362  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:5386 -> 192.168.233.129:80
04/21/2026-20:14:31.645776  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:31511 -> 192.168.233.129:80
04/21/2026-20:14:32.646294  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:52571 -> 192.168.233.129:80
04/21/2026-20:14:33.646098  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:13081 -> 192.168.233.129:80
04/21/2026-20:14:34.646195  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:38805 -> 192.168.233.129:80
04/21/2026-20:14:35.646618  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:61770 -> 192.168.233.129:80
04/21/2026-20:14:36.646958  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:18561 -> 192.168.233.129:80
04/21/2026-20:14:37.645176  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:42480 -> 192.168.233.129:80
04/21/2026-20:14:41.267025  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:61656 -> 192.168.233.129:80
04/21/2026-20:14:42.356345  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:13108 -> 192.168.233.129:80
04/21/2026-20:14:43.356839  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:36926 -> 192.168.233.129:80
04/21/2026-20:14:44.356493  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:5914 -> 192.168.233.129:80
04/21/2026-20:14:45.357376  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:37253 -> 192.168.233.129:80
04/21/2026-20:14:46.357584  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:7180 -> 192.168.233.129:80
04/21/2026-20:14:47.657010  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:7844 -> 192.168.233.129:80
04/21/2026-20:14:48.463331  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:36232 -> 192.168.233.129:80
04/21/2026-20:14:49.463432  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:3530 -> 192.168.233.129:80
04/21/2026-20:14:50.464301  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:32530 -> 192.168.233.129:80
```

8. Recommendations

8.1 Deploy IPS in Critical Systems

Enable inline mode for real-time threat prevention in sensitive environments.

8.2 Strengthen Layer 2 Security

Implement:

- Dynamic ARP Inspection (DAI)
- DHCP Snooping

8.3 Optimize Detection Rules

- Regularly update signatures
- Adjust thresholds for accuracy
- Monitor false positives

```
04/21/2026-20:32:12.393460  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12206 -> 192.168.233.129:80
04/21/2026-20:32:12.393482  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12207 -> 192.168.233.129:80
04/21/2026-20:32:12.393488  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12208 -> 192.168.233.129:80
04/21/2026-20:32:12.393526  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12210 -> 192.168.233.129:80
04/21/2026-20:32:12.393604  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12213 -> 192.168.233.129:80
04/21/2026-20:32:12.393686  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12216 -> 192.168.233.129:80
04/21/2026-20:32:12.393860  [wDrop] [**] [1:1000006:1] Blocked SYN Flood [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12218 -> 192.168.233.129:80
04/21/2026-20:34:10.362203  [**] [1:1000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.233.128:8 -> 192.168.233.129:0
04/21/2026-20:34:21.329268  [**] [1:1000001:1] ICMP Ping Detected [**] [Classification: (null)] [Priority: 3] {IPv6-ICMP} fe80:0000:0000:0000:30b9:99af:af48:de54:143 -> f
04/21/2026-20:38:14.435175  [**] [1:1000003:1] HTTP Traffic Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:58658 -> 172.66.147.243:80
04/21/2026-20:40:13.141977  [**] [1:1000003:1] HTTP Traffic Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.129:50160 -> 92.223.95.95:80
04/21/2026-20:42:28.397716  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:2313 -> 192.168.233.129:80
04/21/2026-20:42:29.401414  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:19146 -> 192.168.233.129:80
04/21/2026-20:42:30.597827  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:24906 -> 192.168.233.129:80
04/21/2026-20:42:31.539404  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:29862 -> 192.168.233.129:80
04/21/2026-20:42:32.989667  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:32007 -> 192.168.233.129:80
04/21/2026-20:42:33.571326  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:33929 -> 192.168.233.129:80
04/21/2026-20:42:34.553777  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:37220 -> 192.168.233.129:80
04/21/2026-20:42:35.561388  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:40638 -> 192.168.233.129:80
04/21/2026-20:42:36.575403  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:50740 -> 192.168.233.129:80
04/21/2026-20:42:37.619781  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:56176 -> 192.168.233.129:80
04/21/2026-20:42:38.615589  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:12855 -> 192.168.233.129:80
04/21/2026-20:42:39.618011  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:29236 -> 192.168.233.129:80
04/21/2026-20:42:40.616642  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:49896 -> 192.168.233.129:80
04/21/2026-20:42:41.616307  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:10214 -> 192.168.233.129:80
04/21/2026-20:42:42.615829  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:40641 -> 192.168.233.129:80
04/21/2026-20:42:43.616811  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:65270 -> 192.168.233.129:80
04/21/2026-20:42:44.619001  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:22044 -> 192.168.233.129:80
04/21/2026-20:42:45.617181  [**] [1:1000004:1] SYN Flood Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.233.128:47563 -> 192.168.233.129:80
```

9. Conclusion

This lab successfully demonstrated the deployment of Suricata as both an IDS and IPS solution. The integration of detection rules, kernel-level filtering, and traffic analysis provides a comprehensive defense mechanism against modern network threats.

The findings highlight the importance of proactive cybersecurity practices and layered defense strategies.